

CVE-2025-51678: An issue was discovered in RISC-V PicoRV32 commit 87c89a. A mismatch in the PCPI INSN and memory address can lead to une

Report Type: Cve Focused · Generated: July 19, 2026 17:10 UTC · Coverage: Jul 17 - Jul 19, 2026

1 Total Advisories	0 Critical	0 High	1 Medium
MEDIUM Severity	N/A CVSS / Risk Score	TLP:CLEAR TLP Classification	PRO Customer Tier

Executive Summary

[P3] CVE / Vulnerability - moderate severity (Risk 5.0/10). No MITRE ATT&CK techniques mapped - limited adversarial context. 1 indicator detected - upgrade to Pro for full IOC access. AI confidence: LOW (22%) - limited source data available.

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Top Threat Advisories

High-priority advisories ranked by CVSS score, exploitability, and actor attribution.

Advisory Title	Severity	CVSS	EPSS	AI Summary
CVE-2025-51678: An issue was discovered in RISC-V	CRITICAL	CVSS: 9.8	16.6%	[P3] CVE / Vulnerability - moderate severity (Risk 5.0/10). No MITR

Vulnerability Intelligence Deep-Dive

N/A CVSS 3.1 Score	16.6% EPSS 30-day Probability	NO CISA KEV Listed	MEDIUM Severity Rating
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Field	Value
CVE / Advisory ID	CVE-2025-51678
Vulnerability Class	Unclassified Security Vulnerability
CWE Reference	Pending NVD enrichment
Actor Attribution	UNC-CDB
Source Advisory	NVD / Vendor Advisory

Refer to the official NVD entry and vendor advisory for the complete list of affected products and versions for CVE-2025-51678.

Vulnerability Context & Attack Surface

This vulnerability affects the security boundary of the target component. Successful exploitation may allow unauthorized access, data disclosure, or denial of service depending on deployment configuration. Consult the NVD entry and vendor advisory for full technical root cause analysis and CVSS vector breakdown.

MITRE ATT&CK® v15 Mapping

Technique density score: 0.0 · Techniques observed: 0 · Tactics covered: 0

MITRE ATT&CK techniques pending analyst enrichment. Deploy detection rules from Section 6 to identify related activity.

Threat Actor Intelligence

Attribution analysis across advisories in this report period.

Threat Actor	Count	Associated CVEs / Indicators
UNC-CDB	1	CVE-2025-51678

Indicators of Compromise (IOCs)

IOC extraction for this advisory is pending enrichment pipeline completion. PRO subscribers receive real-time IOC push to SIEM/SOAR via the SENTINEL APEX webhook at: **GET /api/v1/intel/iocs?advisory={id}**. Full IOC packages include IPv4, domain, SHA256, and URL indicators with confidence scores, first-seen timestamps, and STIX 2.1 formatted bundles.

Detection Engineering

Deploy these production-ready detection rules into your SIEM platform to identify exploitation attempts in real time. Compatible with Microsoft Sentinel, Splunk ES, and any Sigma-compatible platform.

Sigma Rule - Webserver / Process Monitoring

title: SENTINEL APEX - CVE-2025-51678 Exploitation Attempt
id: cdb-cve-2025-51678-det
status: experimental
description: Detects exploitation indicators related to CVE-2025-51678.
references:
- https://intel.cyberdudebivash.com
author: CyberDudeBivash SENTINEL APEX
date: 2026/07/19
tags:
- attack.initial_access
- attack.t1190
logsource:
category: webserver
detection:
keywords:
- 'CVE-2025-51678'
condition: keywords
falsepositives:
- Security testing / penetration testing
level: medium

Microsoft Sentinel - KQL Query

// SENTINEL APEX - CVE-2025-51678: An issue was discovered in RISC-V PicoRV32 commit 87c89a. A mismatch in the PCPI INSN and memory address can lead to une
// CVE-2025-51678 | Microsoft Sentinel KQL
let time_window = ago(24h);
SecurityAlert
| where TimeGenerated > time_window
| where Description has "CVE-2025-51678"
or AlertName has "CVE-2025-51678"
or ExtendedProperties has "CVE-2025-51678"
| extend Rule = "APEX-CVE202551678", Severity = "MEDIUM"
| project TimeGenerated, AlertName, AlertSeverity, Entities, Description, Rule, Severity
| order by TimeGenerated desc

Incident Response Playbook

Phase 1 - 0-24 Hours (Containment)

1. Activate IR team; assign lead analyst and executive sponsor.
2. Isolate or WAF-shield CVE instances exposed to the internet.
3. Enable verbose access logging on all CVE endpoints immediately.

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- 4. Pull last 72h of web server and application logs for forensic baselining.
- 5. Check SENTINEL APEX IOC feeds and SIEM alerts for exploitation hits in your environment.
- 6. Notify relevant internal stakeholders and legal / compliance team.

Phase 2 - 24-72 Hours (Eradication)

- 1. Apply vendor patch for CVE-2025-51678 or implement recommended mitigation.
- 2. Deploy Sigma and KQL detection rules from Section 6 across all SIEM platforms.
- 3. Rotate all API keys, service account credentials, and session tokens.
- 4. Conduct forensic timeline reconstruction for potential compromise window.
- 5. Validate patch deployment across 100% of affected instances via asset inventory.

Phase 3 - 7 Days (Recovery & Hardening)

- 1. Conduct post-incident review and update incident response runbooks.
- 2. Threat hunt for lateral movement or persistence artifacts using MITRE ATT&CK mapping.
- 3. Update asset inventory with patched version information and closure evidence.
- 4. Submit findings to internal risk register and CISO dashboard.
- 5. Schedule follow-up vulnerability scan to confirm remediation completeness.

Financial Impact Analysis (FAIR Model)

\$150K - \$900K Breach Cost Range (FAIR)	\$380K IBM Cost of Breach 2025 Median	\$120K/day Business Interruption Cost	\$85K Regulatory Fine Exposure
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Medium severity events require internal investigation and targeted customer notification. Average cost includes 340 hours of engineering time.

Remediation Cost Estimate: \$55K · Includes engineering time, IR retainer activation, forensic investigation, and customer notification costs.

Regulatory & Compliance Implications

Organizations processing this advisory must evaluate obligations under applicable regulatory frameworks. The table below maps this threat to relevant compliance requirements.

Framework	Reference	Obligation
PCI-DSS v4.0	Req. 6.3.3	Non-critical patches applied within 3 months
NIST CSF 2.0	ID.RA / PR.PS	Vulnerability assessment and platform security policies
ISO 27001:2022	A.8.8	Vulnerability management programme with documented remediation timelines

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Actionable Recommendations

Prioritized defensive actions based on this period's intelligence.

1. Monitor CVE-2025-51678: An issue was discovered in RISC-V PicoRV32 commit 87c89a. A mismatch in the PCPI INSN and memory address can lead to une for escalation.
2. Apply patches in next maintenance window.
3. Apply vendor patch for CVE-2025-51678: An issue was discovered in RISC-V PicoRV32 commit 87c89a. A mismatch in the PCPI INSN and memory address can lead to une immediately - check NVD for official fix guidance.
4. Enable enhanced logging and alerting on all systems running affected software versions.
5. Deploy the Sigma detection rule from this report into your SIEM platform (Sentinel / Splunk / Elastic).
6. Audit all internet-facing instances of affected products and confirm patch deployment.
7. Conduct a threat hunt using MITRE ATT&CK techniques mapped in this advisory.

Appendix - Report Metadata & Legal

Field	Value
Report ID	intel--19496c8d59cdaa83
Report Type	Cve Focused
Platform	CYBERDUDEBIVASH SENTINEL APEX v166.0 GOD-MODE
Generated At	2026-07-19T17:10:14.58498
Customer Tier	PRO
Customer Email	-
TLP	TLP:CLEAR
STIX Version	2.1
ATT&CK Version	v15

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